CAVAL

Clinical AI Validation, Attribution & Lineage

An open standard that defines what must be captured at the moment AI output influences a clinical decision.

Named

Every AI output attributed to source

Traceable

Complete decision chain captured

Defensible

Auditable record for any review

Open Standard
CC BY 4.0

FHIR R4
Aligned

Jurisdiction
Configurable

github.com/cavallayer

CAVAL™

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Clinical AI is deployed. Clinical AI governance is not.

86%

of system-affiliated hospitals using predictive AI (2024)

AHA IT Supplement, ONC, Sept 2025

No attribution record

When AI output influences a clinical decision, there is no structured record of what the AI said, whether it was reliable, or whether the clinician acted on it.

No accountability chain

When something goes wrong and AI was active in the workflow, investigations cannot establish what role AI played or who owned the decision that followed.

No governance standard

There is no open, interoperable standard that defines what clinical AI governance data must be captured, how it should be structured, or what makes it auditable.

The gap between AI deployment and AI governance is the largest unaddressed patient safety gap in digital health.

CAVAL defines the pre-decisional governance layer.

At every point where AI output influences a clinical decision, CAVAL captures a structured attribution event.

AI Output Surfaced

Tool identity
Output content
Model version

Validation Check

In validated state?
Confidence threshold
Drift flag

Clinician Judgment

Accepted
Rejected
Modified / Ignored

Attribution Record

Named owner
Timestamp
Audit trail

FHIR R4 aligned Integrates with existing NHS and US EHR infrastructure without custom connectors

Jurisdiction-configurable Maps to US FDA SaMD, UK CQC, EU AI Act, and Gulf regulatory frameworks

Append-only governance store Tamper-evident audit trail — records cannot be modified after creation

SHA-256 hashed identifiers Patient and clinician identifiers stored as salted hashes — never raw

CAAS v0.1.1 — Clinical AI Attribution Schema

Published at github.com/cavallayer · CC BY 4.0 · JSON Schema

event
Core attribution event — encounter ID, timestamp, AI tool identity, encounter type

tool_activity_log
What the AI tool surfaced — output type, content hash, confidence, model version

provenance_chain
Full lineage — training data version, deployment date, validation status at time of use

review_record
Post-hoc review — reviewer identity, findings, escalation status

decision_context
Clinical context at decision point — specialty, workflow, risk level, urgency

judgment_capture
Clinician action — accepted / rejected / modified / ignored, with rationale

governance_metadata
Regulatory mapping — jurisdiction, applicable framework, compliance tags

promotion_conditions
Threshold definitions — what conditions must be met for AI output to be acted on

Full schema: github.com/cavallayer/cavallayer/blob/main/schema/caas-v0.1.1.json

Where CAVAL applies in clinical AI workflows.

Ambient Documentation

Who: Ambient scribe vendors, telehealth platforms

Captures whether AI-generated clinical note content was accepted, modified, or overridden by the clinician before signing

Establishes clinician ownership of the final record and surfaces where AI documentation diverges from clinical judgment at scale

Diagnostic Decision Support

Who: Radiology AI, pathology AI, clinical decision support vendors

Records what AI recommendation was surfaced, whether the model was in a validated state, and what the clinician decided

Creates the audit trail needed when an AI-assisted diagnosis is challenged in a malpractice or regulatory review

Prior Authorization AI

Who: Payers, utilization management platforms

Documents what AI output was used in the authorization decision and what threshold triggered approval or denial

Satisfies CMS requirements for AI-assisted prior auth transparency and provides defensible records for appeals

Risk Stratification

Who: Population health platforms, care management tools

Captures which risk score influenced which care intervention and whether the model had drifted from its validated state

Enables post-hoc analysis of whether AI-guided interventions were based on reliable outputs — critical for value-based contracts

How CAVAL fits into your architecture.

What the open standard defines (free)

- The CAAS JSON schema — all required fields and data types
- The event structure — what must be captured at each decision point
- Jurisdiction mapping guidance — US, UK, EU, Gulf
- FHIR R4 alignment specifications
- The governance roadmap — what is coming in v0.2 and beyond

What production implementation requires

- A conformance-validated governance store — the append-only database that captures CAVAL events
- A validation engine — checks each attribution event against the schema before writing
- Integration with your AI tool's output layer — typically an API call at decision point
- Jurisdiction configuration — which regulatory framework applies to your deployment context

Production implementation tooling available through Kairoc Systems · kairoc.ai

Where CAVAL is heading.

NOW — v0.1

Foundation

- CAAS v0.1.1 schema published (CC BY 4.0)
- Core attribution event structure defined
- FHIR R4 alignment specified
- Jurisdiction-configurable framework established
- GitHub repository live and public

NEAR TERM — v0.2

Validation

- Reference implementation and validator
- Conformance testing tooling
- Pilot implementations with design partners
- Evidence generation from real clinical workflows
- Schema refinement based on pilot feedback

MID TERM — v1.0

Certification

- Conformance certification program
- Multi-jurisdiction regulatory recognition
- EHR vendor integration specifications
- Clinical AI governance store reference builds
- International design partner network

Roadmap is directional. Timing reflects current development priorities and may change.